Requirements Table — Dropshipping Inventory & Order Orchestrator

Functional Requirements

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	**Pass:** Supplier PO triggered with customer delivery address. **Fail:** Order placed for out-of-stock supplier SKU.	Automating PO creation removes manual order entry and prevents the store owner from unknowingly selling stock the supplier no longer has.
FR-002	The system shall split a single customer order containing SKUs from multiple supplier partners into distinct per-supplier sub-orders, each retaining the original delivery address and a shared customer order reference.	High	**Pass:** An order with SKUs from 2 different suppliers produces 2 separate supplier POs, both tagged with the same parent order ID. **Fail:** A multi-vendor order is sent as a single combined PO to only one supplier, or line items are dropped during the split.	Suppliers can only fulfill items in their own catalog, so multi-vendor orders must be decomposed or some items will never ship.
FR-003	The system shall query each relevant supplier's inventory API in real time before confirming line-item availability back to the storefront.	High	**Pass:** The system confirms "in stock" only after a live API call made within the current processing session. **Fail:** Availability is confirmed using cached data older than the defined freshness threshold (e.g., > 5 min).	Selling against stale stock data causes cancellations and damages the store's reputation with end customers.
FR-004	The system shall ingest tracking numbers pushed from supplier APIs/webhooks and propagate them to the corresponding storefront order.	Medium	**Pass:** A tracking number appears on the storefront order within 5 minutes of the supplier update. **Fail:** Tracking data is received by the system but never reflected on the customer-facing order.	Customers expect shipment visibility without the store owner having to manually relay supplier tracking info.

ID	Description	Priority	Acceptance Criteria	Rationale
FR-005	The system shall notify the E-Store Owner when a supplier partially fulfills, rejects, or backorders any part of a split order.	Medium	**Pass:** The E-Store Owner receives a dashboard/email notification within 2 minutes of a supplier rejection or backorder event. **Fail:** An order segment remains unfulfilled with no notification sent to the owner.	Store owners need timely visibility to intervene (refund, re-source, notify the customer) before delivery SLAs are breached.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storms.	High	**Pass:** Benchmarking tests confirm target latency and security standards under simulated peak load. **Fail:** Acknowledgement exceeds 200 ms under load, triggering provider retries.	Shopify/WooCommerce aggressively retry unacknowledged webhooks; slow acks cause duplicate deliveries and cascading load.
NFR-002	Reliability (Idempotency)	The system shall guarantee idempotent processing of webhook and PO-dispatch events, such that retried webhook deliveries never result in duplicate purchase orders for the same order line item.	High	**Pass:** Sending the same webhook payload 5 times in a load test results in exactly one PO per line item. **Fail:** Any duplicate PO is created for the same order/line-item combination.	Because NFR-001 relies on fast acknowledgement, retries are expected behavior — without idempotency guarantees, retries would create duplicate supplier orders and billing errors.